Jungbae Park (Bobby) | Resume

- **Status:** ML Engineer @ Bucketplace (오늘의집) · S.Korea
- **Links:** Portfolio Link · Linkedin
- **Objective:** ML Engineer, Research Scientist, Entrepreneur
- **Interests:** AI Agent · Retrieval/RecSys · GenAI · ML/LLMOps · Multimodal Representation · Healthcare · HCI · NLP · Speech/Audio · Neuroscience-inspired AI

*"I believe in the brightening and cozy future, led by **cognitive computing**." To pursue this vision, I love to learn, develop, and research—bridging AI agents, retrieval/RecSys, GenAI, ML/LLMOps, multimodal learning, healthcare, HCI, and neuroscience-inspired approaches to build production systems that make a real-world impact. Besides R&D, I enjoy creating something novel: inventing, prototyping, mentoring, and contributing to society through technology.*

Work Experiences

→ See detailed per-project outcomes on p.2–4

01/2023 – Present | ML Engineer (Search & AIL; Applied Intelligence Lab) | Bucketplace (오늘의집)

- **AIL**: Led ML-side development and optimization of GenAI systems—OHouseAI workflow (**Net Satisfaction Score +253%**, latency **-50%**); contributed to scene-to-products retrieval for Digital Twin (**13×** recall, enabling **6000×** cost reduction in image-to-3D pipeline); now building context/intent-aware agentic natural language compositional retrieval/ranking for RoomPlanner; also built a multi-agent R&D workflow automation plugin (17 agents, 14 skills, cross-model review).
- **Search**: Architected and delivered search ranking and multimodal retrieval—shipped **11** A/B experiments (**8 winners**) across Store SRP (SERP) and Category PLP (CPLP): multi-objective HPO (SERP CTR **+0.95%**, CPLP purchase/query **+1.62%**; **CIKM'25** oral, patent applied), deals ranking (buyer conv up to **+11%**), and CLIP-based hybrid retrieval for recall boost (Query CTR **+3.03%**, CTCVR **+16%**); built ML exp. infra (offline evaluation, log monitoring, automated A/B analysis) and shared ML Mart. (~2023.07 전문연구요원)

07/2020 – 01/2023 | ML Engineer, Research Scientist (MLOps & AI Research) | RIIID (뤼이드)

- **MLOps**: Shipped and operated model registry, dataset pipelines, and distributed training infrastructure across **4+ EdTech products** (SANTA TOEIC, IVYGlobal SAT, CASA GRANDE, INICIE); improved GPU utilization **25% → 95%** by introducing the company's first multi-GPU / data-distributed training setup.
- **AI Research**: Achieved **SOTA** on student modeling (dropout prediction, conditional dropout prediction, knowledge tracing) via attentive conditional contrastive learning (ACCL) across 6 benchmarks; developed and deployed multimodal (text+audio+image) speech scoring system to SANTA Say (TOEIC Speaking), leveraging cross-modal context for cold-start robustness; published at **InterSpeech 2023**. (전문연구요원)

04/2018 – 06/2020 | Research Lead; COO; Co-Founder | Humelo (휴멜로)

- **Research**: Managed and developed **5+** AI R&D projects—Emotional TTS (**ICASSP 2019** oral), Voice Conversion (**ICASSP 2020**), Speech Emotion Recognition, Lyrics Generation, and Sound Event Detection (**ICASSP 2019**)—publishing **3 top-venue papers** as 1st/corresponding author.
- **Management**: Co-founded and operated as COO (HR, finance, culture); secured **~$875K** across **3** government R&D grants (IITP, TIPS, Seoul R&BD); grew company from founding through Series A stage.

Education

03/2019 – 08/2022 | Ph.D. Student (Lab for Brain & Machine Intelligence) | KAIST

- Withdrawal due to startup co-foundation & military service. (Advisor: Prof. Sang Wan Lee)

03/2017 – 02/2019 | Master of Science (Lab for Brain & Machine Intelligence) | KAIST

- in Bio and Brain Engineering (Advisor: Prof. Sang Wan Lee)
- Master's Thesis: Attentional Control Methods for Time-series Data Classification and Synthesis (2019)

02/2012 – 02/2017 | Bachelor of Science | KAIST

- in Bio and Brain Engineering, Minor in Computer Science (CS) (Excellent in Volunteering & Leadership)

03/2009 – 02/2012 | High School Diploma (Div. of Natural Sciences) | Chungnam High School

- Korean SAT mock ranking: top 21% (2009.03) → top 1.x% (2011.06) (nation's percentile).

Tech Stack & Languages

Programming	Python (primary), SQL · also able to handle: Kotlin, Go, C++, Java, JavaScript, Scala, Dart, MATLAB, R, Shell
ML/DL & AutoML	PyTorch, TensorFlow, HuggingFace, Lightning AI, XGBoost, LightGBM, LoRA, Optuna, Ray Tune
MLOps & Serving	BentoML, Triton Inference Server, TorchServe, MLFlow, Airflow (K8s), W&B
Agent Frameworks	LangChain, LangGraph, Agent Development Kit (ADK), LangFuse
Data & Search Infra	Spark (PySpark/Scala), ElasticSearch, Athena, BigQuery, Redis, PostgreSQL, MongoDB, Milvus
Web/App	FastAPI, Django REST, React, Flutter
Cloud & CI/CD	AWS (EC2, S3, ECR, RDS, Lambda), GCP (Cloud Computing, GCS), Docker, GitHub Actions
AI Coding / Protocol	Claude Code, Codex, Cursor, MCP, Skills, A2A
Collaboration	Jira, Notion
Languages	Korean (Native), English (Business, TOEIC 890; 2020)
Extra Specialties	Music Production (Logic Pro, Ableton Live; Mixing/Mastering), DJing, Inventing, Volunteering

Projects

01/2026 – Current | **Agentic Multi-modal Natural Language 3d Model Search** | Bucketplace (오늘의집)

- Building an AI-powered search agent that interprets natural language queries into structured retrieval signals (category, attributes, color, dimensions, budget) via LLM, executing BM25+KNN hybrid search on ElasticSearch for Room Planner 3D products.
- Designed **Col-Fit** (Context-Intent Fit Matching)—a compositional multimodal retrieval framework combining space analysis, mood/style, dimensional constraints, and conversational context from image + text + 3D coordinate inputs—serving as the retrieval backbone for multiple downstream agents.
- Architected a **LangGraph** pipeline with parallel fan-out inference, auto quality recovery via filter relaxation retry, dual interface (A2A JSON-RPC + REST/FastAPI), and persona-based evaluation pipeline generating synthetic query-document sets to evaluate retrieval relevance across long-tail distributions.

LangGraph / A2A / ADK / LangFuse / ElasticSearch / Airflow / Triton / FastAPI / VLM / Multimodal Retrieval

01/2026 – Current | **rnd-quality-helper: Multi-Agent R&D Workflow & Code Review Automation** | Bucketplace (오늘의집)

- Built a Claude Code plugin (team-level PoC) with **17** specialized agents and **14** skills automating code review, planning, and verification across the dev lifecycle (Research → Plan → Implement → Verify → Iterate).
- Implemented cross-model multi-persona review (Claude/Codex/Gemini in parallel), 3-layer defense (Hook → Command → CI), mode switching (research/production), and Notion integration—addressing self-preference bias through multi-LLM consensus and ReAct feedback loops.

Claude Code Plugin / Multi-Agent Orchestration / Cross-Model Review / LLM / ReAct / Notion MCP

12/2025 – 01/2026 | **Scene-to-Products Retrieval for Digital Twin (Image-to-3D)** | Bucketplace (오늘의집)

- Reduced lead time **2 weeks → 3 days** and improved **Recall@10 2.06% → 28.08% (13×+)** by building E2E segmentation → indexing → demo validation and implementing a process-centric gen AI pipeline (refine → representative extraction → validate/filter → indexing).
- Removed **56.7%** defective images (**>2× yield**), scaled **30k → >400k** products across **154 → 277 categories (+80%)**; SAM3D pipeline achieved **~1/100** cost vs outsourced ImgTo3D, **6000× cost reduction** and **2200× production increase** with 3D collaboration.

3D Reconstruction / Segmentation / Object Detection / GenAI Pipeline / Vector Indexing

07/2025 – 11/2025 | **OHouseAI GenAI Workflow Systematization** | Bucketplace (오늘의집)

- Improved service metrics (**Net Satisfaction Score +253%**, **PN +205%**; requests/user **×2.3**; retention **7% → 14%**; latency **71.7s → 35.7s**) by modularizing GenAI workflow (pipeline provider + subgraph). Outperformed GPT-IMAGE-1.5 / Gemini Nano (**8.3/10** intent, **8.1/10** context).
- Built LLM-as-a-Judge batch evaluation pipeline (**4 rounds × 99 sets**)—automated side-by-side comparisons with structured scoring (intent reflection, context preservation) for data-driven model selection.

LangChain / LangGraph / LangFuse / LLM-as-a-Judge / Pipeline Provider

02/2025 – 06/2025 | **Search OKR: E-commerce Deals / Price 2.0 Ranking** | Bucketplace (오늘의집)

- Delivered measurable OKR impact through production experiments (**4 total; 3 winners**) by establishing a reusable experimentation foundation (Query Feature Table → Redis → Server) and shipping PRD/Design Doc → DAG → prod rollout end-to-end; developed specialized Grid Search HPO.
- STORE SRP deals Buyer Conv **+1.99%**, Click Conv **+0.83%**, special-offer exposure **+7.35%**; Category PLP Buyer Conv **+11.17%**, Click Conv **+5.87%**, exposure **+27.9%**; minimal side effects.

Redis / Airflow DAG / Grid Search HPO / Optuna / Ray Tune / ElasticSearch

10/2023 – 01/2025	**MOHPER: Multi-Objective HPO for E-commerce Search Ranking**	Bucketplace (오늘의집)

➧ Built and deployed MOHPER, a multi-objective HPO framework addressing selection bias and data sparsity in e-commerce retrieval—launched **>5×** tuned params on SERP and **>3×** on CPLP (online: SERP CTR **+0.95%**, Click/Query **+1.23%**; CPLP CTCVR **+1.10%**, Purchase/Query **+1.62%**); accepted to **CIKM'25** Applied Science Track (oral) & patent applied; expanded into AD domain & common Search Team stack.

Optuna / Ray Tune / ElasticSearch / Hydra-core / Bayesian Optimization

07/2023 – 09/2023	**Global Content Search Ranking**	Bucketplace (오늘의집)

➧ Built two-stage content ranking (candidate retrieval + function-score reranking) for 3 content types with time decay and global language analyzer; Redis-cached serving.

ElasticSearch / Airflow / Go Server

04/2023 – 06/2023	**Search ML Infra & Experiment Automation**	Bucketplace (오늘의집)

➧ Designed shared ML feature mart for Search Team; improved data quality monitoring via client-side log validation and lag anomaly detection; automated A/B test analysis pipeline for experiment metric computation and reporting.

PySpark / HiveDB / Airflow

01/2023 – 06/2023	**CLIP-based E-Commerce Downstream Tasks Design & R&D**	Bucketplace (오늘의집)

➧ Shipped CLIP-based hybrid retrieval on Store SRP for recall boost via text-to-image vector search (Query CTR **+3.03%**, CTCVR **+16.39%**) with scalar quantization (**2× speed**, **1/4 disk**); classification Top10 Acc **92.23%** among >2000 categories.

CLIP / Quantization / Multimodal Retrieval / HNSW / ElasticSearch/ ANN

10/2022 – 01/2023	**Multimodal Prompt-aware Speech Scoring System R&D**	RIIID (뤼이드)

➧ Proposed prompt embeddings + question context (BERT/CLIP) to address cold-start item problem in E2E speech scoring; deployed to SANTA Say (TOEIC Speaking); accepted to **InterSpeech 2023**.

Speech Scoring / Cold-start / TOEIC / PyTorch

08/2021 – 10/2022	**User Modeling: Knowledge Tracing & Dropout Prediction**	RIIID (뤼이드)

➧ Achieved **SOTA** on student modeling (dropout prediction, conditional dropout prediction, knowledge tracing) with proposed attentive conditional contrastive learning (ACCL)—trainable attentional coefficients reweighting CL objectives by conditional input similarity—across 6 benchmarks; deployed to Santa TOEIC via BentoML.

Contrastive Learning / BentoML / Knowledge Tracing / PyTorch / Transformer

01/2021 – 09/2021	**ML Model Deploy, Registry & Dataset Pipeline**	RIIID (뤼이드)

➧ Built model registry (MLFlow) and dataset pipelines (Airflow, Athena, BigQuery) serving **4+ products** (SANTA TOEIC, IVYGlobal SAT, CASA GRANDE, INICIE).

MLFlow / Apache Airflow / AWS Athena / GCP BigQuery / Docker

07/2020 – 12/2020	**ML Training Pipeline Optimization & Research Engineering**	RIIID (뤼이드)

➧ Introduced first multi-GPU training in the company & improved all pipeline procedures (GPU utilization **25% → 95%**; init time **1 hr → 10 sec**); refactored common modules into efficient batch-style preprocessing for atypical log-tabular data; contributed ML pre/post-processing and user modeling to personalized content recommendation for SANTA TOEIC.

Multi-GPU / Data-Distributed Training / CI/CD

06/2019 – 03/2020	**Emotional Voice Conversion**	Humelo (휴멜로)

➧ Managed and co-developed multi-speaker, multi-domain emotional voice conversion using Factorized Hierarchical VAE for disentangled emotion representations; enabled conversion across multiple emotion types without voice distortion. Published at **ICASSP 2020**.

Disentangled Representation / VAE / Voice Conversion / TensorFlow

04/2018 – 03/2020	**Government & Public R&D Grants**	Humelo (휴멜로)

➧ Planned and led R&D projects from government & public organizations:

- IITP research team grant: Development of brain-inspired AI (No.2019-0-01371), 2019–2020. Role: co-PI, **~US$225K**
- Ministry of SMEs/startups R&D: Next-gen deep-learning emotion/expression TTS (S2644149), TIPS, 2018–2020. Role: Leading researcher; co-manager; co-proposal-author, **~US$400K**
- Seoul R&BD Program: Voice Conversion (VC) and TTS for VC (CY190019), 2019–2020. Role: Leading researcher; co-manager; co-proposal-author, **~US$250K**

04/2018 – 03/2020	**Duration Controllable TTS & Emotional TTS**	Humelo (휴멜로) & KAIST

› Managed and developed phonemic-level duration control (PDC) for end-to-end speech synthesis using teacher attention alignment; achieved controllable speech speed without quality loss. Published as **oral** at **ICASSP 2019** (1st author).

E2E TTS / Neural Vocoder / Attention Alignment / TensorFlow / Speech Synthesis

06/2018 – 05/2019	**Polyphonic Sound Event Detection**	Humelo (휴멜로)

› Managed and developed convolutional bidirectional LSTM with synthetic data-based transfer learning; F1 score **+28.4%** and error rate **-0.42** on TUT 2016 dataset. Published at **ICASSP 2019** (corresponding author).

Transfer Learning / TensorFlow

04/2018 – 03/2019	**Composing, Lyrics Generation AI & Rapper AI**	Humelo (휴멜로) & KAIST

› Managed and developed LSTM-based chord progression melody composing (ALPAHCH), lyrics generation, and rap synthesis system. Showcased at **SXSW 2019**; featured in **KBS Documentary** with professional rapper Sleepy; covered by **10+ national media** outlets.

Seq2Seq / Music Generation / NLP / TensorFlow

04/2018 – 02/2019	**Speech Emotion Recognition & Classification**	Humelo (휴멜로)

› Managed and developed speech emotion classification system for the Emotional TTS pipeline; built feature extraction and classifier for multi-class emotion detection from audio signals.

SpeechCNN, CRNN / MFCC + Mel Analysis / Emotion Recognition / TensorFlow

06/2017 – 02/2019	**Adaptive EEG Signal Processing with Deep RL**	KAIST

› Developed deep RL-based attentional control for EEG time-series classification, solving the memory-based vs. memoryless trade-off problem; published as **oral** at **IEEE SMC 2018**; patent registered (KR). Funded by IITP (No. 2017-0-00451).

Deep RL / BCI (EEG) / Time-series Classification / TensorFlow / MATLAB / Lasagne

Personal Projects

01/2024 – Present	**AI Group Calendar Service**

› Polymorphic RecSys engine with LLM, vector-embedding, and feature-based backends for intelligent scheduling; dual-queue async processing (LLM vs. ML latency tiers); peer-pattern learning via opt-in role-model calendars.

Flutter / Django REST / PostgreSQL / pgvector / Firebase / Redis / LangGraph / Celery / Fly.io

01/2023 – Present	**AI Quant Test**

› Multi-strategy quantitative analysis platform with configurable signal generation (momentum, mean-reversion, multi-timeframe trend); automated daily ingestion pipeline and indicator-driven backtesting framework.

Django REST / Airflow / MariaDB / PyTorch / scikit-learn

Academic Publications (* = corresponding author)

› *Jungbae Park, Heonseok Jang – **MOHPER: Multi-objective Hyperparameter Optimization Framework for E-commerce Retrieval System** (Applied Science Track, to appear). **CIKM 2025** – Top Conference on Information Retrieval & Data Management

› *Jungbae Park, *Seungtaek Choi – **Addressing Cold Start Problem for End-to-end Automatic Speech Scoring**. **Interspeech 2023** – Top Conference on Audio & Signal Processing

› *Jungbae Park, Soonwoo Kwon, Jinyoung Kim, Sang Wan Lee – **SACCL: Sequential User Representations with Relation Contrastive Learning** (See portfolio).

› *Jungbae Park, et al. – **SAICL: Student Modelling with Interaction-level Auxiliary Contrastive Tasks for Knowledge Tracing and Dropout Prediction**. **arXiv 2023**

› Hyunmook Park, *Jungbae Park, *Sang Wan Lee – **End-to-end Trainable Self-Attentive Shallow Network for Text-Independent Speaker Verification**. **arXiv 2020**

› Mohamed Elgaar, *Jungbae Park, *Sang Wan Lee – **Multi-speaker and Multi-domain Emotional Voice Conversion Using Factorized Hierarchical VAE**. **ICASSP 2020** – Top Conference on Audio & Signal Processing

› Jungbae Park et al. – **Phonemic-level Duration Control Using Attention Alignment for Natural Speech Synthesis** (oral). **ICASSP 2019** – Top Conference on Audio & Signal Processing

› Seokwon Jeong, *Jungbae Park, *Sang Wan Lee – **Polyphonic Sound Event Detection Using Convolutional Bidirectional LSTM and Synthetic Data-based Transfer Learning**. **ICASSP 2019** – Top Conference on Audio & Signal Processing

- Jungbae Park, *Sang Wan Lee – **Solving the Memory-based-Memoryless Trade-off Problem for EEG Signal Classification** (oral). **IEEE SMC 2018**
- Jungbae Park, Jaryong Lee, *Sang Wan Lee – **A New Approach for LSTM Polynomial Melody Composition based on Finite Chord Progression**. **KIIS 2017**
- Jungbae Park, Juno Kim, *Sang Wan Lee – **Multi-agent Cognitive Policy Learning: Reinforcement Learning Through Competition** (**Best Paper Award**). **KIIS 2016**

Patents (8 Registered, 3 Applied)

- **Registered (>1 nation):**
  I. Adaptive EEG signal processing using reinforcement learning – KR (1023187750000). Jungbae Park, Sang Wan Lee
  II. Voice conversion system and method – KR (1022772050000). Mohamed Elgaar, Jungbae Park
  III. System/device/method to generate polyphonic music – KR (1022274150000) + PCT. Jungbae Park, Jaryong Lee
  IV. Training sound event detection model – KR (1021724750000). Seokwon Jeong, Jungbae Park
  V. Training sound event detection model – KR (1020256520000) + PCT. Seokwon Jeong, Jungbae Park
  VI. Apparatus for synthesizing speech – KR (1020579270000) + PCT. Jaryong Lee, Jungbae Park
  VII. Apparatus for synthesizing speech – KR (1020579260000) + PCT. Jungbae Park, Kijong Han
  VIII. Vibration sensor – KR (1012602650000, ended). Jungbae Park
- **Applied/Applying:**
  I. E-commerce retrieval support – KR (10-2025-0027980) / US / JP. Jungbae Park, Heonseok Jang
  II. Supervised contrastive learning for patch-level temporal representation – KR (1020230026429). Jungbae Park
  III. Training sound event detection model – KR (1020200139661). Seokwon Jeong, Jungbae Park

Awards (selected)

- **Special Award, Minister Award of Science and ICT** (특별상, 과기정통부 장관상): Challenge! K-Startup (2018) – (Team: Humelo; Jaryong Lee, Jungbae Park, Yongseok Kwan, Seokwon Jeong, Yuneui Jeong)
- **3rd Position, Excellent Awards** (우수상) | Organizer: Skill-up; Supervisor: Blue Point Partners (2018): Lab-based R&D Industrialization Accelerating Program, "Lab2Biz" (Team: Humelo; Jungbae Park, Jaryong Lee, Yongseok Kwan)
- **Excellent Awards, KAIST Alumni President Awards** (우수상, KAIST 동문회장상): KAIST Start-up Competition (2018) – (Team Name: Humelo; Jungbae Park, Jaryong Lee, Yongseok Kwan, Seokwon Jeong, Yuneui Jeong)
- **Certificate of Diamond Level** KAIST Leadership Program (2017)
- **Outstanding Paper Award** 2016 Fall Korea Intelligent Information System Conference (2016)
- **Excellent Awards, Minister Award of Education** (우수상, 교육인적자원부 장관상): National Science Exhibition < Section of Plant-related Research > (2011)
- **Gold Prize** for 2011 Daejeon Invention Contest (Superintendent of Education Awards, 교육감상, 2011)
- **Silver Prize** for 2010 Daejeon Inventions Idea Contest (Superintendent of Education Awards, 교육장상, 2010)

Activities & Presentations (selected)

- **12.2021 – 01.2022: Part-time Online Deep Learning Lecturer @FastCampus** (패스트캠퍼스). Lecture: (KR) 한 번에 끝내는 딥러닝 / 인공지능 초 격차 패키지 Online (Part 5 강의): Data science best seller on 2022 first half & Steady seller among all domains (from 03.2022). Also able to handle: (KR) 입문자를 위한 딥러닝 정주행 Kit (Step 2 강의).
- **11.2021 – PyCon KR 2021 – Full-session (30 mins) Presenter.** Title: Can AI models be deployed "without code modification" in multiple domains with one codebase & pipeline? (하나의 코드 베이스 & 파이프라인으로 여러 도메인에 "코드 수정 없이" AI 모델들을 배포할 수 있을까?) Oral Presentation Video. PPT Material.
- **05.2020 – EduKOCCA Online Lecture Seminar @Korea Creative Content Agency (KOCCA).** Lectures: (KR) 4차 산업혁명, AI, 딥 러닝, 그리고 뉴 콘텐츠 – AI 를 적용한 사운드 콘텐츠 알아보기; AI 를 음악 프로듀싱에 어떻게 이용할 수 있을까?
- **03.2019 – SXSW 2019 Presentation:** AI-aided Music Production by Rapper AI & Composing AI: XXX – Flight Attendant (Integrating Sound and Visual Interpretation). Collaborated with Artists, 'XXX' (Korean Hip-hop Band), 'Seonggu de Kim' (Sound Designer, DJ), and 'Seong Pil Kim' (Visual Artist).
- **02.2019** – Introduced "Rapper AI" and "Composing/Arranging AI" on KBS1 Documentary, "Documentary-world: New-contents, changing the world)" (다큐 세상 – 세상을 바꾸는 뉴콘텐츠, 1st, Feb, KBS1). AI rearranged * featuring? Rapper Sleepy and AI collaboration rap revealed. Video.
- **11.2017** – Leader of ART X AI team, Play with Error. Creative Innovation Project 1st Music X A.I. with SM Ent. Music, turns on AI, by KOCCA. Development of Composing AI with Jaryong Lee and Create AI-aided Music and Visual arts with Artists. Related Articles.
- **06.2017 ~ 03.2018** – Graduate School Student Researcher: BCI(EEG) + AI (especially using Deep RL) research, funded by IITP (No. 2017-0-00451).

Extra Activities

- **Undergraduate Research** (2015–2017): Brain & Machine Intelligence (RL), Neuro-Machine Augmented Intelligence (EMG), Music & Audio Computing (Composing AI), Brain Dynamics (Connectome analysis).

- **Teaching**: KAIST TA (3 courses: Brain Functions, Bio-data Structure, Brain-inspired AI); Counseling Assistant; Private math tutor (2014–2016).
- **Clubs**: Student Council planning officer, Global Volunteering Council (founded Daejeon network), Game Dev (HAJE), DJing (FUZE), Composing (LP), Inventing (KAINOVATOR).